

Recon-ng

Objective

The objective of this task is to perform footprinting and reconnaissance on a target domain using the Recon-ng framework. The task aims to gather publicly available information such as hosts, subdomains, domains, and contact details using automated reconnaissance modules.

This activity helps in understanding:

- Open-Source Intelligence (OSINT) gathering
- Reconnaissance techniques used in cybersecurity
- Usage of Recon-ng modules and workspaces
- Automated information gathering methods
- Analysis of publicly available target information

The main goals are:

1. To learn the working of Recon-ng.
2. To perform domain and host enumeration.
3. To collect publicly available target information.
4. To organize reconnaissance data using workspaces.
5. To understand the importance of reconnaissance in ethical hacking and penetration testing.

Recon-ng Commands with Simple Explanation

1. Start Recon-ng: recon-ng

This command opens the Recon-ng tool.

2. Create a Workspace: workspaces create fortune_workspace

A workspace is like a folder/project inside Recon-ng.

This command:

- Creates a new project named fortune_workspace
- Saves all reconnaissance data inside it

Example:

If you scan multiple websites, each website can have its own workspace.

3. View All Workspaces: workspaces list

Shows all created workspaces.

Example output:

default
fortune_workspace

4. Search Available Modules: marketplace search

Recon-ng uses small tools called modules.

This command:

- Displays all available modules
- Helps find modules for reconnaissance

Think of modules as plugins/features.

5. Search Domain Modules: marketplace search domains-hosts

Shows modules related to:

- Domains
- Hosts
- Subdomains

Used for website/domain reconnaissance.

6. Install a Module: marketplace install recon/domains-hosts/bing_domain_web

Downloads and installs the Bing search module.

This module:

- Uses Bing search engine
- Finds subdomains and hosts related to a website

Example:

If target is example.com, it may find:

- mail.example.com
- admin.example.com

7. Load the Module: modules load recon/domains-hosts/bing_domain_web

Opens the installed module so you can use it.

After loading:

- Module becomes active
- You can configure settings

8. View Module Settings: options list

Shows the settings/options required for the module.

Example:

SOURCE
LIMIT

SOURCE = target website/domain

9. Set the Target Website: options set SOURCE example.com

Sets the target domain.

Replace example.com with your target website.

Example: options set SOURCE google.com

10. Run the Module: run

What this command does

Starts reconnaissance.

The module now:

- Searches the internet
- Collects subdomains
- Finds hosts related to the target

Example results:

mail.example.com
dev.example.com

11. Show Found Hosts: show hosts

Displays all discovered hosts/subdomains.

Example:

mail.example.com
admin.example.com

These are systems related to the target domain.

12. Load WHOIS Module: modules load recon/domains-contacts/whois_pocs

Loads a module that collects WHOIS contact information.

WHOIS information may include:

- Email addresses
- Domain owner details
- Administrative contacts

13. Set Target Again: options set SOURCE example.com

Specifies which website/domain to investigate.

14. Run WHOIS Module: run

Starts WHOIS information gathering.

The module searches public domain registration databases.

15. Show Collected Contacts: show contacts

Displays gathered contact information.

Example:

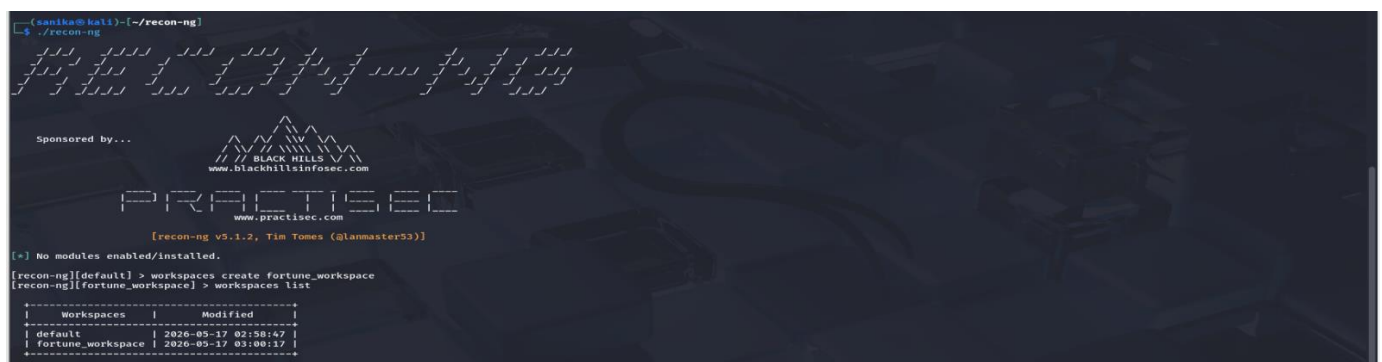
```
admin@example.com  
support@example.com
```

16. Exit Recon-ng: exit

Closes Recon-ng safely.

All gathered data stays saved in the workspace.

Screenshot



```
(sanika@kali)~[/recon-ng]  
$ ./recon-ng  
  
Sponsored by...  
www.blackhillsinfosec.com  
www.practiseec.com  
[recon-ng v5.1.2, Tim Tones (@lanmaster53)]  
[*] No modules enabled/installed.  
[recon-ng][default] > workspaces create fortune_workspace  
[recon-ng][fortune_workspace] > workspaces list  


| Workspaces        | Modified            |
|-------------------|---------------------|
| default           | 2020-05-17 02:50:47 |
| fortune_workspace | 2020-05-17 03:00:17 |


```

```
[recon-ng][default] > workspaces create fortune_workspace
[recon-ng][fortune_workspace] > workspaces list

+-----+-----+
| Workspaces | Modified |
+-----+-----+
| default    | 2026-05-17 02:58:47 |
| fortune_workspace | 2026-05-17 03:00:17 |
+-----+-----+

[recon-ng][fortune_workspace] > workspaces load fortune_workspace
[recon-ng][fortune_workspace] > workspaces remove fortune_workspace
[recon-ng][default] > marketplace search

+-----+-----+-----+-----+-----+-----+
| Path | Version | Status | Updated | D | K |
+-----+-----+-----+-----+-----+-----+
| recon/discovery/info_disclosure/cache_snoop | 1.1 | not installed | 2020-10-13 | | |
| recon/discovery/info_disclosure/interesting_files | 1.2 | not installed | 2021-10-06 | | |
| recon/exploitation/injection/command_injector | 1.0 | not installed | 2019-06-26 | | |
| recon/exploitation/injection/paths_bruter | 1.2 | not installed | 2019-10-08 | | |
| recon/import/csv_file | 1.1 | not installed | 2019-08-09 | | |
| recon/import/list | 1.1 | not installed | 2019-06-26 | | |
| recon/import/masscan | 1.0 | not installed | 2020-04-07 | | |
| recon/import/nmap | 1.1 | not installed | 2020-10-06 | | |
| recon/companies-contacts/bing_linkedin_cache | 1.0 | not installed | 2019-06-26 | | * |
| recon/companies-contacts/censys_email_addresses | 2.1 | not installed | 2022-01-31 | | * |
| recon/companies-contacts/pen | 1.1 | not installed | 2019-10-15 | | |
| recon/companies-domains/censys_subdomains | 2.1 | not installed | 2022-01-31 | | * |
| recon/companies-domains/pen | 1.1 | not installed | 2019-10-15 | | |
| recon/companies-domains/viewdns_reverse_whois | 1.1 | not installed | 2021-08-24 | | |
| recon/companies-domains/whois_dns | 1.1 | not installed | 2020-06-17 | | * |
| recon/companies-multi/censys_org | 2.1 | not installed | 2022-01-31 | | * |
| recon/companies-multi/censys_tls_subjects | 2.1 | not installed | 2022-01-31 | | * |
| recon/companies-multi/github_miner | 1.1 | not installed | 2020-05-15 | | * |
| recon/companies-multi/shodan_org | 1.1 | not installed | 2020-07-01 | | * |
| recon/companies-multi/whois_miner | 1.1 | not installed | 2019-10-15 | | |
| recon/contacts-contacts/abx | 1.0 | not installed | 2019-10-11 | | * |
| recon/contacts-contacts/mailtester | 1.0 | not installed | 2019-06-26 | | |
| recon/contacts-contacts/mangle | 1.0 | not installed | 2019-06-26 | | |
| recon/contacts-contacts/umangle | 1.1 | not installed | 2019-10-27 | | |
| recon/contacts-credentials/hulp_breach | 1.2 | not installed | 2019-09-10 | | * |
| recon/contacts-credentials/hulp_paste | 1.1 | not installed | 2019-09-10 | | * |
| recon/contacts-domains/censys_email_to_domains | 2.1 | not installed | 2022-01-31 | | * |
| recon/contacts-domains/migrate_contacts | 1.1 | not installed | 2020-05-17 | | |
| recon/contacts-profiles/fullcontact | 1.1 | not installed | 2019-07-26 | | |
| recon/credentials-credentials/adobe | 1.0 | not installed | 2019-06-26 | | |
| recon/credentials-credentials/bozerack | 1.0 | not installed | 2019-06-26 | | |
| recon/credentials-credentials/hashes_org | 1.0 | not installed | 2019-06-26 | | |
| recon/credentials-credentials/whois_computer | 1.1 | not installed | 2021-06-11 | | |
```

```
D = Has dependencies. See info for details.
K = Requires keys. See info for details.

[recon-ng][default] > marketplace search domains-hosts
*) Searching module index for 'domains-hosts'...

+-----+-----+-----+-----+-----+-----+
| Path | Version | Status | Updated | D | K |
+-----+-----+-----+-----+-----+-----+
| recon/domains-hosts/binaryedge | 1.2 | not installed | 2020-06-18 | | * |
| recon/domains-hosts/bing_domain_api | 1.0 | not installed | 2019-06-26 | | * |
| recon/domains-hosts/bing_domain_web | 1.1 | not installed | 2019-07-06 | | |
| recon/domains-hosts/brite_hosts | 1.0 | not installed | 2019-06-26 | | |
| recon/domains-hosts/builtwith | 1.1 | not installed | 2021-08-24 | | * |
| recon/domains-hosts/censys_domain | 2.1 | not installed | 2022-01-31 | | * |
| recon/domains-hosts/certificate_transparency | 1.1 | not installed | 2019-09-16 | | |
| recon/domains-hosts/google_site_web | 1.0 | not installed | 2019-06-26 | | |
| recon/domains-hosts/hackertarget | 1.1 | not installed | 2020-05-17 | | |
| recon/domains-hosts/mx_spf_ip | 1.0 | not installed | 2019-06-26 | | |
| recon/domains-hosts/netcraft | 1.1 | not installed | 2020-02-05 | | * |
| recon/domains-hosts/shodan_hostname | 1.1 | not installed | 2020-07-01 | | * |
| recon/domains-hosts/srsr_subdomains | 1.1 | not installed | 2021-08-26 | | * |
| recon/domains-hosts/ssl_san | 1.0 | not installed | 2019-06-26 | | |
| recon/domains-hosts/threatcrowd | 1.0 | not installed | 2019-06-26 | | |
| recon/domains-hosts/threatminer | 1.0 | not installed | 2019-06-26 | | |

+-----+-----+-----+-----+-----+-----+

D = Has dependencies. See info for details.
K = Requires keys. See info for details.

[recon-ng][default] > marketplace install recon/domains-hosts/bing_domain_web
*) Module installed: recon/domains-hosts/bing_domain_web
*) Reloading modules...
[recon-ng][default] > modules load recon/domains-hosts/bing_domain_web
[recon-ng][default][bing_domain_web] > options list

Name      Current Value Required Description
-----
SOURCE    default      yes      source of input (see 'info' for details)
```

```

Name      Current Value Required Description
-----
SOURCE    default      yes      source of input (see 'info' for details)

[recon-ng][default][bing_domain_web] > SOURCE
+-----+
[recon-ng][default][bing_domain_web] > LIMIT
+-----+
[recon-ng][default][bing_domain_web] > SOURCE LIMIT
+-----+
*) You can command SOURCE LIMIT
[recon-ng][default][bing_domain_web] > options set SOURCE testasp.vulnweb.com
SOURCE => testasp.vulnweb.com
[recon-ng][default][bing_domain_web] > options set SOURCE google.com
SOURCE => google.com
[recon-ng][default][bing_domain_web] > run

-----
GOOGLE.COM
-----
*) URL: https://www.bing.com/search?first=0&q=domain3Agoogle.com
[recon-ng][default][bing_domain_web] > show hosts
*) No data returned.
[recon-ng][default][bing_domain_web] > modules load recon/domains-contacts/whois_pocs

[recon-ng][default][bing_domain_web] >
[recon-ng][default] > modules load recon/domains-contacts/whois_pocs

[recon-ng][default] > show contacts
*) No data returned.
[recon-ng][default] >
```